Compiled Problems – Intermediate Division

1. **01-02 C2 Bit String Flicking**

   Evaluate:

   (RSHIFT−2 (RCIRC−8 (RSHIFT−2 10011)))

2. **01-02 C2 Bit String Flicking**

   How many values of X (5 bits long) satisfy the following equation?

   (RSHIFT−1 X) OR 10110 AND 00101 = 00101

3. **02-03 C2 Bit String Flicking**

   Simplify the following expression:

   (LCIRC−2 (NOT 10110) OR (RSHIFT−2 (LCIRC−1 10100 AND NOT 00001)))

4. **02-03 C2 Bit String Flicking**

   Evaluate $A27_{16}$ XOR $4E2_{16}$

   The two arguments are hexadecimal numbers representing bit strings that are 12 bits long. Express your answer as a 3-digit hexadecimal string.

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5. **03-04 C2 Bit String Flicking**

   Evaluate the following expression.

   (NOT (RSHIFT-1 (RCIRC-2 (NOT (LSHIFT-1 01101)))))

6. **03-04 C2 Bit String Flicking**

   How many different values of X (a 5-bit string) satisfy the following expression.

   RCIRC-3 10101 OR LCIRC-2 X = 10111

7. **04-05 C2 Bit String Flicking**

   Evaluate:

   (RCIRC-3 (LSHIFT-1 (LCIRC-2 (RSHIFT-1 10110))))

8. **04-05 C2 Bit String Flicking**

   Solve for X (a 5-bit string):

   (LCIRC-2 01110) AND X = (RSHIFT-1 10011)

9. **05-06 C2 Bit String Flicking**

   A new operation CAT-S,L is defined as follows: Concatenate the bits starting in location S with a length of L to the right-hand side of the original string. Evaluate:

   (RCIRC - 3 (LSHIFT – 4 (CAT – 2,3 0110111)))

10. **05-06 C2 Bit String Flicking**

    Evaluate:

    (RCIRC–2 (LSHIFT–1 (NOT 01101))) OR (RSHIFT-3 (LCIRC-2 11100))

11. **06-07 C2 Bit String Flicking**

    Evaluate:

    ((RSHIFT-2 (10110 AND 10010)) OR (LCIRC-3 (01101 OR 10001)))

12. **06-07 C2 Bit String Flicking**

    List all the values of X (a 5-bit string) that make the following expression TRUE:

    (LSHIFT-2 (RCIRC-1 (NOT X))) = 10000

13. **07-08 C2 Bit String Flicking**

    Evaluate the following expression:

    (NOT 10111 OR LCIRC-2 01110 OR LCIRC-1 00110 AND RSHIFT-2 10110)

14. **07-08 C2 Bit String Flicking**

    List all possible values of X (5 bits long) that solve the following equation:

    (LSHIFT-2 01011 AND RCIRC-2 X) = 00100

15. **08-09 C2 Bit String Flicking**

Evaluate the following expression:

(RCIRC-3 10110) OR (NOT 01110) XOR (LSHIFT-1 01101)

16. **08-09 C2 Bit String Flicking**

List all possible values of X (5 bits long) that solve the following equation:

(LSHIFT-1 10111) AND X = (RSHIFT-1 (RCIRC-2 10010))

17. **09-10 C2 Bit String Flicking**

Evaluate the following expression:

(LCIRC-2 (RSHIFT-1 01100)) AND (LSHIFT-1 (RCIRC-3 10011))

18. **09-10 C2 Bit String Flicking**

List all possible values of X (5 bits long) that solve the following equation:

(LCIRC-2 (RSHIFT-1 X)) OR (LSHIFT-1 (RCIRC-2 00011)) = 11001

19. **10-11 C2 Bit String Flicking**

Evaluate the following expression:

(RCIRC-2 (LSHIFT-1 (10011 AND NOT 00110)))

20. **10-11 C2 Bit String Flicking**

List all possible values of X (5 bits long) that solve the following equation:

(LCIRC-3 (RSHIFT-2 01110)) OR (RCIRC-2 X) = 11011

21. **11-12 C2 Bit String Flicking**

Evaluate the following expression:

(LSHIFT-2 00100) OR (RCIRC-2 01101) AND (NOT 01100)

22. **11-12 C2 Bit String Flicking**

Evaluate the following expression:

(LCIRC-2 (NOT 11011)) OR (RSHIFT-1 01101)

23. **12-13 C2 Bit String Flicking**

Evaluate the following:

NOT 10110 OR 11001 AND 01110

24. **12-13 C2 Bit String Flicking**

Evaluate the following:

(RSHIFT-2 01101) AND (LCIRC-1 01100) OR (NOT 11011)

25. **13-14 C2 Bit String Flicking**

Evaluate the following expression:

(NOT (10010 OR NOT 10011 AND 11011))

26. **13-14 C2 Bit String Flicking**

Evaluate the following expression:

((LSHIFT–2 01101) OR 01011 AND (RCIRC–3 11011))

27. **14-15 C2 Bit String Flicking**

Evaluate the following expression:

10101 OR (NOT 01001) AND (10110 OR 01110)

28. **14-15 C2 Bit String Flicking**

Evaluate the following expression:

(LCIRC-2 (RSHIFT-1 01101)) OR (NOT (RCIRC-2 (LSHIFT-1 01010)))

29. **15-16 C2 Bit String Flicking**

Evaluate the following expression:

(LSHIFT-1 (RCIRC-2 (RSHIFT-1 (LCIRC-3 01101))))

30. **15-16 C2 Bit String Flicking**

Evaluate the following expression:

(LCIRC-2 (RSHIFT-1 01101)) AND ((RCIRC-1 01100) OR (NOT 01101))